# Cedarbridge IT Ticketing Platform — Project Acceptance Criteria

## AC-01: Complete End-to-End Ticket Management

The system must support the entire IT ticket lifecycle, from initial submission through resolution and closure.

**Acceptance conditions:**

- Employees can create tickets and receive unique reference numbers.
- Tickets are persistently stored with the required information, timestamps, and current status.
- Authorized IT staff can categorize, prioritize, assign, reassign, escalate, and resolve tickets.
- The backend enforces valid status transitions and prevents unauthorized or invalid updates.
- Important ticket changes are recorded with timestamps, responsible identities, and relevant previous/new values.
- Closed tickets remain available for authorized historical retrieval.

**Validation:** Demonstrate a ticket moving through its complete lifecycle, including reassignment, reopening, and an invalid status-change attempt. Confirm the record survives an application restart.

## AC-02: Secure Identity, Authorization & Data Protection

The system must protect internal IT information and enforce appropriate access based on user responsibilities.

**Acceptance conditions:**

- Users authenticate using an approved identity mechanism, preferably Microsoft Entra ID.
- Employees can access only tickets they are authorized to view.
- IT support staff can manage permitted requests based on their assigned role.
- Authorization is enforced by the backend, not only by hiding frontend buttons.
- Service-to-service authentication and sensitive application configuration are managed securely.
- Unauthorized access attempts are rejected and relevant security events are recorded.

**Validation:** Test employee and IT support accounts, attempt prohibited ticket access and updates, and verify that credentials are not exposed in the repository or application responses.

## AC-03: Employee Transparency & Management Visibility

The system must provide employees, IT staff, and leadership with appropriate visibility into ticket progress and operational performance.

**Acceptance conditions:**

- Employees can view current ticket status, assignment information where appropriate, and resolution details.
- Employees receive submission confirmation and relevant status notifications.
- IT staff can view incoming, assigned, unresolved, and aging tickets.
- Management can view basic operational reporting, including outstanding requests, request categories, and resolution trends.
- Reports reflect the underlying ticket records accurately.
- Notifications and reporting respect role-based access restrictions.

**Validation:** Submit, update, and resolve multiple synthetic tickets, then verify that notifications, employee views, IT queues, and management reports accurately reflect the changes.

## AC-04: Reliable Backend Processing, Error Handling & Recovery

The system must process requests consistently, maintain data integrity, and provide sufficient diagnostics to investigate failures.

**Acceptance conditions:**

- The backend validates incoming data and returns appropriate responses for invalid requests.
- Concurrent or repeated operations do not unintentionally corrupt ticket records or create duplicate actions.
- Application and database failures produce controlled errors rather than silent data loss.
- Relevant application events, exceptions, and dependencies are logged.
- Application health can be monitored and failures investigated.
- A documented, practical recovery procedure is tested successfully.

**Validation:** Simulate invalid inputs, concurrent updates, repeated requests, and an unavailable dependency. Verify expected application behaviour, inspect diagnostic logs, and complete a recovery exercise.

## AC-05: Reproducible Azure Deployment & Operational Management

The complete solution must be deployable and manageable through documented, repeatable engineering processes.

**Acceptance conditions:**

- Required Azure infrastructure is defined and provisioned using Terraform.
- Application code and infrastructure configurations are maintained in GitHub.
- A CI/CD pipeline automatically validates code changes and supports controlled application deployment.
- Cloud access and deployment credentials follow least-privilege security practices.
- The deployed application functions correctly through its Azure-hosted endpoint.
- Infrastructure costs can be estimated, monitored, and controlled.
- Deployment, troubleshooting, recovery, and resource cleanup procedures are documented.

**Validation:** Provision the environment using Terraform, deploy the application, run end-to-end tests, inspect monitoring information, and demonstrate a controlled update and cleanup procedure.

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# Final Project Acceptance

The project will be considered technically complete when all five criteria have passed their documented validation tests and the system can demonstrate a secure, functional ticket lifecycle in Azure.

Formal production readiness would additionally require stakeholder approval of operational responsibilities, security controls, support arrangements, and any remaining business policies.